

FATE P05, P06

Algorithmic Fairness II

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*** Slide content based on the material made by Ashwin Singh (ashwin.singh@tuwien.ac.at) ***



Universitat
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Fairness, Accountability, Transparency
and Ethics of Data Processing (FATE)

Programming Sessions



1. **6 Labs = 3 Modules x 2 Sessions**

2. **Deliverables per Module**

A zip file containing the notebook (worth 10 points) submitted through Aula Global

3. **Grading Policy**

~~M1: Data Anonymization (35%) Submit by 11:59 PM, Feb 02, 2026~~

~~M2: Algorithmic Fairness I (30%) Submit by 11:59 PM, Feb 19, 2026~~

M3: Algorithmic Fairness II (35%) Submit by 11:59 PM, Mar 10, 2026

4. **Queries / Discussion**

Please post on the Aula Global Forum for everyone's benefit.

5. **Solved Practices Update:** Will not be available, refer to tips and guidance on github.

Outline



In **Module II**, we will focus on **Classification**

1. Evaluating Classifiers: Confusion Matrices, ROC & EDC Curves.
2. Pre / In / Post - Processing Interventions to Improve Fairness Properties.
3. How to do (2) using [AI Fairness 360](#) by IBM Research.
4. Project [**You!**]

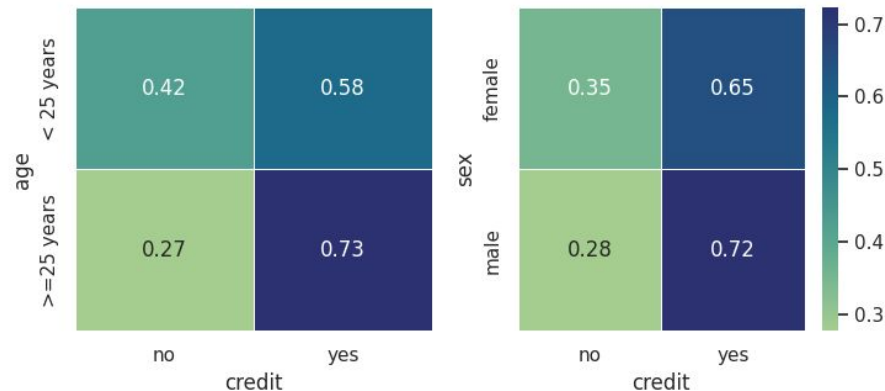
Datasets

1. German Credit Scoring

German Credit Scoring Data

age	sex	X_1	X_2		X_n	credit	%
<=25	male	...	...	...	...	no	38
		...	...	...	...	yes	61
	female	...	...	...	...	no	45
		...	...	...	...	yes	55
>=25	male	...	...	...	...	no	26
		...	...	...	...	yes	74
	female	...	...	...	...	no	30
		...	...	...	...	yes	70

$A = \{ \text{sex, age} \}$ $X = \{ X_1, X_2, \dots, X_n \}$ $Y = \text{credit}$



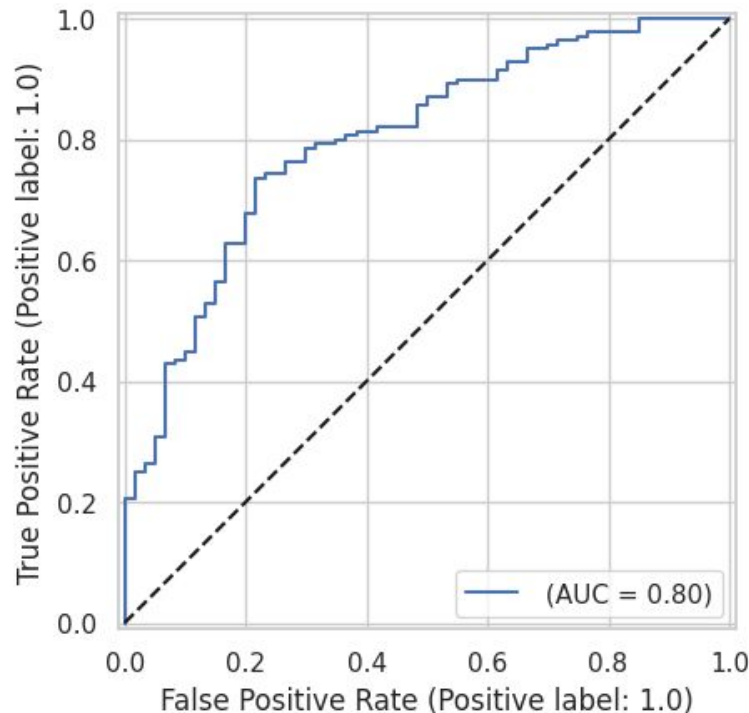
Evaluating Classifiers ► ROC Curves

		Predictions	
		$\hat{Y} = 0$	$\hat{Y} = 1$
Ground Truth	$Y = 0$	TN	FP
	$Y = 1$	FN	TP

False Positive Rate = $FP / (FP + TN)$

True Positive Rate = $TP / (TP + FN)$

Each point in the **ROC Curve** corresponds to a classification threshold θ such that $h(x) > \theta \Rightarrow \hat{Y} = 1$.



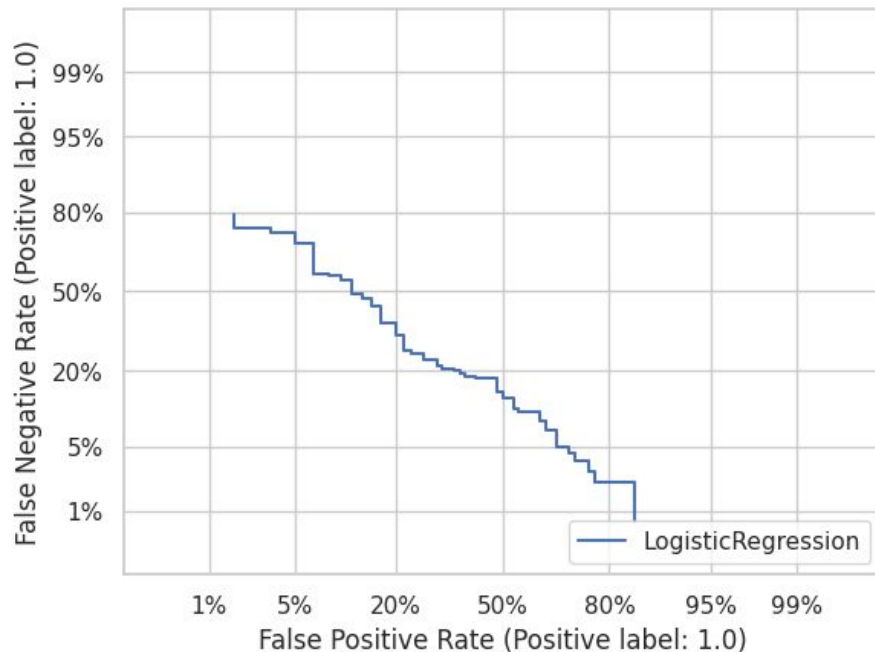
Evaluating Classifiers ► EDC Curves

		Predictions	
		$\hat{Y} = 0$	$\hat{Y} = 1$
Ground Truth	$Y = 0$	TN	FP
	$Y = 1$	FN	TP

False Positive Rate = $FP / (FP + TN)$

False Negative Rate = $FN / (TP + FN)$

Each point in the **EDC Curve** corresponds to a classification threshold θ such that $h(x) > \theta \Rightarrow \hat{Y} = 1$.



Fairness Metrics ► Adapted to Loan Approval

Loan approval is an **assistive** intervention.

Therefore, focus is on protected category **false negatives**.

At the same time, the **approval rate** across privileged and unprivileged groups by must be similar.

Disparate Impact

$$\frac{P(\hat{Y} = 1 \mid A = \text{unprivileged})}{P(\hat{Y} = 1 \mid A = \text{privileged})}$$

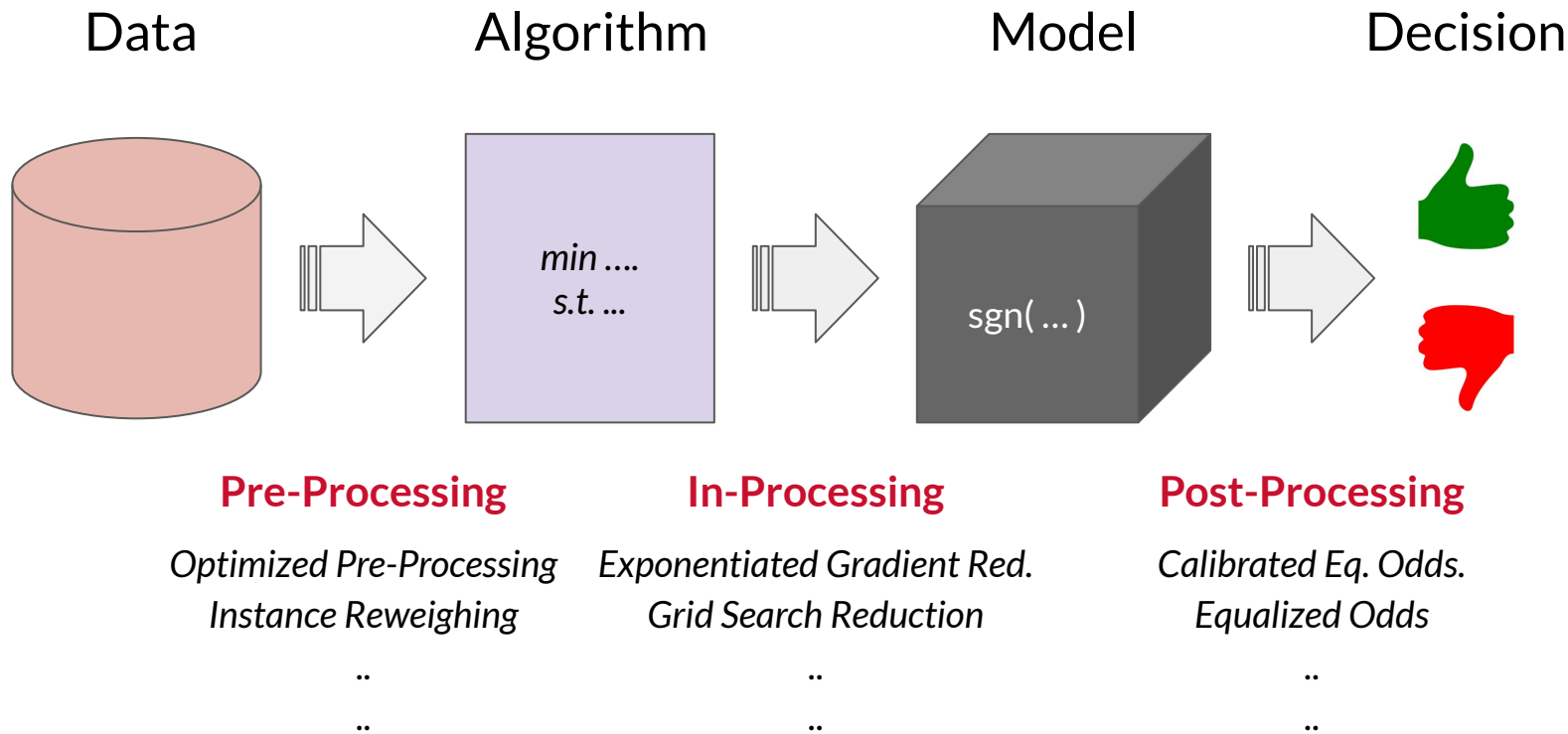
False Negative Rate Ratio

$$\frac{FNR_{A=\text{unprivileged}}}{FNR_{A=\text{privileged}}}$$

False Negative Rate Difference

$$FNR_{A=\text{unprivileged}} - FNR_{A=\text{privileged}}$$

Interventions for Fairness



Interventions for Fairness ► Pre-Processing

Optimized Pre-Processing transforms Data $A, X, Y \Rightarrow A, X', Y'$ such that

1. The dependence of Y' on A is bounded. For any two groups $a_1, a_2 \in A$, the difference in dependencies of Y' on a_1 and a_2 is also bounded.
2. Expected pointwise distortions in the transformation are minimal.
3. Differences (KL-divergence) in the probability distribution underlying the data before and after the transformation are bounded.

Instance Re-Weighting re-weights each sample in data to ensure $Y \perp A$.

$$P(Y = 1, A = a) = P(Y = 1) \times P(A = a) \text{ and vice versa for } Y = 0.$$

Exponentiated Gradient Reduction

Trains the classifier subject to constraints towards ensuring one of **demographic parity** / **equalized odds** for all combinations of protected attributes in data.

Recall that a classifier h satisfies:

Demographic Parity $[h \perp A]$

$$P[h(x) = \hat{y} \mid A = a] = P[h(x) = \hat{y}] \quad \forall a \in A$$

Equalized Odds $[h \perp A \mid Y]$

$$P[h(x) = \hat{y} \mid A = a, Y = y] = P[h(x) = \hat{y} \mid Y = y] \quad \forall a \in A, y \in Y$$

Calibrated Odds-Equalizing

Optimizes over calibrated classifier (h) score outputs using a linear program to find probabilities with which to change output labels to satisfy:

Equalized Odds $[h \perp A | Y]$

$$P[h(x) = \hat{y} | A = a, Y = y] = P[h(x) = \hat{y} | Y = y] \quad \forall a \in A, y \in Y$$

AI Fairness 360



These are ten state-of-the-art bias mitigation algorithms that can address bias throughout AI systems. Add more!

Optimized Pre-processing

Use to mitigate bias in training data. Modifies training data features and labels.



Reweighting

Use to mitigate bias in training data. Modifies the weights of different training examples.



Adversarial Debiasing

Use to mitigate bias in classifiers. Uses adversarial techniques to maximize accuracy and reduce evidence of protected attributes in predictions.



Reject Option Classification

Use to mitigate bias in predictions. Changes predictions from a classifier to make them fairer.



Disparate Impact Remover

Use to mitigate bias in training data. Edits feature values to improve group fairness.



Learning Fair Representations

Use to mitigate bias in training data. Learns fair representations by obfuscating information about protected attributes.



Prejudice Remover

Use to mitigate bias in classifiers. Adds a discrimination-aware regularization term to the learning objective.



Calibrated Equalized Odds Post-processing

Use to mitigate bias in predictions. Optimizes over calibrated classifier score outputs that lead to fair output labels.



Equalized Odds Post-processing

Use to mitigate bias in predictions. Modifies the predicted labels using an optimization scheme to make predictions fairer.



Meta Fair Classifier

Use to mitigate bias in classifier. Meta algorithm

Algorithms

- `aif360.algorithms.preprocessing`
- `aif360.algorithms.inprocessing`
- `aif360.algorithms.postprocessing`

Project ► Datasets

Dataset	Protected Category	Group
Adult - <u>aif360.datasets.AdultDataset</u> <i>Task: Predict whether annual income of an individual exceeds \$50K/year based on census data.</i>	Sex { male / female }	
	Race { white / non-white }	
	Native-Country { national / immigrant }	
ProPublica Compas - <u>aif360.datasets.Compas</u> <i>Task: Predict whether an individual will re-offend</i>	Sex { male / female }	
	Race { caucasian / other }	

Project ► Datasets

Dataset	Protected Category	Group
Default of Credit Card Clients - ucimlrepo (id=350) <i>Task: Predict whether a customer will face the default situation in the next month.</i>	Sex { male / female }	
	Marital Status { single / married }	
Law School Bar Exam Passage - Course Github <i>Task: Predict whether an applicant will pass the bar exam in first try (as a proxy for admission).</i>	Sex { male / female }	
	Race { white / non-white }	

Project ► Datasets

Dataset	Protected Category	Group
Covid-19 Open Data Mexico - Course Github <i>Task 1: Predict whether an intubated patient will get admitted into intensive care (Intubated patients in ICU have a higher mortality rate).</i>	Sex { male / female }	
	Age { < 60 / >= 60 }	

Tip: if your dataset is too large for processing, you can take a smaller (stratified) sample